WORKED SOLUTION

STEP III 2003, Question 11

Mechanics • Projectile motion

STEP III 2003, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/03-s3-q11

The key idea

Forget the motion for a moment and think about the single point where the collision would happen. At that instant A, B and the collision point form a triangle whose sides are d, the distance P has travelled, and the distance Q has travelled, with the angle at B fixed by the bearing. The cosine rule then gives one equation, and writing both travelled distances in terms of the time turns it into a quadratic. Whether a collision is possible is then a question about whether that quadratic has a positive root.

Solution

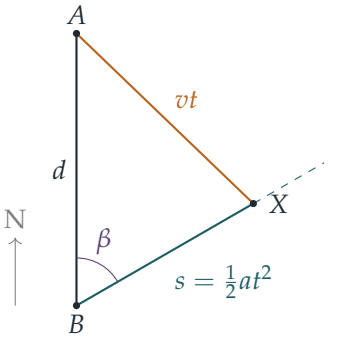
We model P and Q as particles moving in the horizontal plane. Bearings are measured clockwise from north, and $v > 0$.

The collision equation

Suppose Q strikes P at time $t > 0$, at the point X. By then P has started from rest with constant acceleration a along a fixed line, so

$$BX = s = \tfrac{1}{2}at^2,$$

and Q has moved in a straight line at constant speed, so $AX = vt$.



The direction from B to A is due north and the direction from B to X has bearing β, so the angle ABX is β (or $360^\circ - \beta$ if $\beta > 180^\circ$, which has the same cosine). The cosine rule in triangle ABX gives

$$v^2t^2 = d^2 + s^2 - 2ds\cos\beta.$$

(If $\beta = 0$ or 180° the "triangle" is flat, but the equation still holds: it becomes $(vt)^2 = (d \mp s)^2$.) Since $t^2 = 2s/a$, the left-hand side is $2v^2s/a$, and rearranging,

$$s^2 - 2\left(d\cos\beta + \frac{v^2}{a}\right)s + d^2 = 0. \qquad (*)$$

Conversely, if $(*)$ has a root $s > 0$, let $t = \sqrt{2s/a}$ and let X be the position of P at that time. Then $(*)$ says exactly that $AX = vt$, and $AX > 0$. So if Q is projected from A towards X, it reaches X at time t, just as P does, and strikes it. Hence

$$\text{a suitable direction exists} \iff (*) \text{ has a positive root } s.$$

The necessary condition

Write $k = d\cos\beta + v^2/a$, so that $(*)$ is $s^2 - 2ks + d^2 = 0$.

> **Ask yourself** The quadratic certainly needs real roots, but is that enough? What else must we know about them?

The roots must also be *positive*, because s is a distance travelled. The product of the roots is $d^2 > 0$, so the two roots have the same sign, and their sum is $2k$. So there is a positive root exactly when the discriminant is non-negative and $k > 0$:

$$k^2 \geq d^2 \quad \text{and} \quad k > 0 \iff k \geq d.$$

(If $k > 0$ then $k^2 \geq d^2$ is the same as $k \geq d$, as $d > 0$; and $k \geq d$ forces $k > 0$.)

So if Q can strike P, then

$$d\cos\beta + \frac{v^2}{a} \geq d, \qquad \text{that is,} \qquad v^2 \geq ad(1 - \cos\beta),$$

which is the required result.

Striking before P has moved a distance d

Now suppose $v^2 > ad(1-\cos\beta)$, so that $k > d > 0$. The discriminant $4(k^2 - d^2)$ is then strictly positive, and $(*)$ has two distinct roots $s_1 < s_2$. Both are positive, since their sum $2k$ and their product d^2 are positive. As they are distinct with

$$s_1 s_2 = d^2,$$

they can't both be at least d or both at most d, so

$$s_1 < d < s_2.$$

Explicitly, $s_1 = k - \sqrt{k^2 - d^2}$.

By the converse above, projecting Q towards the point X_1 at distance s_1 from B along P's path makes Q strike P at time $t_1 = \sqrt{2s_1/a}$, and at that moment P has moved only $s_1 < d$. So Q can be aimed to strike P before P has moved a distance d, as required.

Checks

- Take $\beta = 0$: P runs straight towards A, so they must meet whatever the value of v. The condition gives $v^2 \geq 0$, which always holds. Then $(*)$ becomes $s^2 - 2(d + v^2/a)s + d^2 = 0$, and the smaller root is below d: they meet before P reaches A, as common sense says.
- Take $a = d = 1$, $\beta = 90^\circ$ (so P runs due east), and $v^2 = 2 > 1 = ad(1-\cos\beta)$. Then $(*)$ is $s^2 - 4s + 1 = 0$, so $s_1 = 2 - \sqrt{3} \approx 0.268 < 1$. At $t_1 = \sqrt{2s_1}$ we have $s_1^2 + 1 = 2t_1^2$, so the collision point really is at distance vt_1 from A. A numerical check over many random values of a, d, β and v confirmed both the hit at $s_1 < d$ and, when $v^2 < ad(1-\cos\beta)$, that P always stays further than vt from A.
- In the borderline case $v^2 = ad(1-\cos\beta)$ we get $k = d$ and a double root $s = d$: the only possible hit is exactly when P has travelled d, which is why the strict inequality is needed in the final part.

Common slips

- Demanding only that the discriminant be non-negative. That gives $|k| \geq d$, which also allows $k \leq -d$, where both roots are negative and no collision happens.
- Using the wrong angle in the cosine rule. The angle at B is between BA (due north) and BX (bearing β), so it is β itself, not $180^\circ - \beta$.
- Only proving the necessary condition in the final part. You must also show that the aim towards X_1 actually works, which is the converse argument.

Beyond STEP

This is a pursuit-and-interception problem. Here the target's path is known in advance and the interceptor moves in a straight line, so the question reduces to a single equation in the meeting time. Missile guidance and "pursuit curves", where the chaser keeps turning towards the target, lead to differential equations instead. The habit of fixing the meeting point first and asking who can reach it in time is a standard tool in first-year mechanics. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops the kinematics of relative motion carefully.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.